

# YUXUAN HOU

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## Education

### Southern University of Science and Technology (SUSTech)

Expected Jun 2028

*B.Eng. in Computer Science and Technology — Turing Class (Honors Program)*

Shenzhen, China

**Academic-year GPA: 3.86/4.00** | **Class President**, 2024 CSE Turing Class

**Honors Coursework:** Discrete Mathematics; Data Structures and Algorithm Analysis; Algorithm Design and Analysis; Principles of Database Systems; Artificial Intelligence.

**Other Completed Coursework:** Computer Organization; Computer Vision; Natural Language Processing; Digital Logic; Introduction to Mathematical Logic; Introduction to Computer Programming; Computational Ethics; Group Projects I.

**In Progress:** Operating Systems (Honors); Compiler Principles; Computer Networks; Advanced Computer Programming; Probability and Statistics for Engineering.

## Research Affiliation

**Lab:** Advanced Intelligent Mobile Systems (AIMS) Lab, SUSTech

**Advisor:** Jin Zhang, Associate Professor, Department of Computer Science and Engineering

## Accepted Publication

Yuxuan Hou, Jiao Li, Linshan Jiang, and Jin Zhang. *Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness*.

**WellComp 2026 (UbiComp/ISWC 2026 Workshop)** — Computing for Well-being.

Accepted; forthcoming in *UbiComp Companion '26*. DOI: 10.1145/3798063.3841767. arXiv:2608.16678.

Additional manuscripts are in preparation.

## Research Experience

### Oto-Meal: Personalized Earable Sensing

Sep 2025–Present

*Project Lead & Primary Developer*

*Wearable sensing*

**Project Lead** — 2026 National Undergraduate Innovation and Entrepreneurship Training Program (**National Level**; CNY 25,000)

**Project Lead** — 2026 SUSTech “Climbing Program” Project (CNY 15,000; completed)

- Built an audio- and image-free photoplethysmography (PPG) and inertial measurement unit (IMU) pipeline with two-stage behavior recognition and user-conditioned calibration; evaluated **538 recordings and 7,028 windows from 7 participants**.
- Achieved  **$80.38 \pm 0.84\%$**  five-class event accuracy with 20% labeled target-user calibration under the paper’s window-split protocol; led the accepted first-author manuscript and continue follow-up modeling and evaluation.

### Neuromorphic Sensing: Event Encoding for Wireless Signals

Ongoing

*Method Development & Experimental Evaluation*

*In preparation*

- Develop spiking neural network (SNN) encoders, including frequency-budgeted events, membrane-state feedback, and Haar-based gated encoding; build comparisons across **nine datasets**, with component ablations and protocol-specific accuracy and computation analysis.

### ReActLens: State-Aware Tool-Result Delivery

Ongoing

*Systems Prototyping & Evaluation*

*In preparation*

- Prototype tool-result delivery coordinated with cache residency and inference scheduling for resource-constrained language-model agents; evaluate public tool-use traces with a real inference engine, including a **112-run, 1,792-request** controlled study of recovery latency.

Research Experience (continued)

3DGS-Acoustic: Transferable Acoustic Object Responses

May 2026–Present

Acoustic Modeling & Simulation

In preparation

- Progressed from DiffRIR reproduction to compact object-response transfer for scene editing; developed simulation comparisons and repeated-edit cost evaluation, using **26.67% fewer measurement records** than the specified recalibration baseline over four edits, with lower mean change error in that setting.

InterveneCompose: Structural Property Testing

Ongoing

Theoretical Analysis & Computational Verification

In preparation

- Investigate query complexity for testing structural properties of multi-label functions; develop lower-bound constructions and an exact test under a second-order structural assumption, supported by proofs and exact-arithmetic checks on **180 instances**.

Q-CalTrack: Phase-Based Motion Recovery

Ongoing

Signal Processing & Error Analysis

In preparation

- Develop and diagnose phase-based trajectory recovery with sparse anchors; correct frequency-axis and motion-removal errors, and analyze residual blind spots, phase ambiguity, and conditional trajectory uncertainty on existing recordings.

Selected Engineering Projects

**Object-Level 3D Gaussian Splatting** (Spring 2026; custom-scene lead, team of four): built reconstruction, segmentation, and editing workflow; registered 213/213 frames with 0.71-pixel reprojection error.

**RydborgTrack** (May–Jun 2026; signal processing, team of three): built a reproducible MATLAB pipeline for 150 million samples; separated 3 static and 12 motion recordings within the project dataset.

Technical Skills

**Programming:** Python, C/C++, MATLAB, Java, Verilog

**Learning & Scientific Computing:** PyTorch, scikit-learn, NumPy, SciPy

**Systems & Research Tools:** CUDA, COLMAP, Vivado, Git, Linux, LaTeX; experimental evaluation, literature review, and bilingual research communication

Selected Honors & Awards

|         |                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025    | <b>National First Prize</b> , Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)                                                        |
| 2025    | <b>Silver Medal (Rank 4)</b> , International Collegiate Programming Contest (ICPC)                                                                       |
| 2025    | <b>Silver Medal</b> , China Collegiate Programming Contest (CCPC)                                                                                        |
| 2025    | <b>Gold Medal</b> , Guangdong Collegiate Programming Contest                                                                                             |
| 2026    | <b>First Prize (1st Place)</b> , Application Development Division Finals, 1st Hong Kong International AIGC Cultural Digital Content Creation Competition |
| 2025    | Provincial First Prize, 16th Lanqiao Cup, Guangdong; C/C++ Programming, University Group A                                                               |
| 2024/25 | SUSTech University Scholarship; Zhiren College Motto Scholarship “Rixin”; SUSTech Outstanding Student Leader                                             |
| 2024    | Undergraduate Talent Development Excellence Program Grant (2024 Cohort; CNY 6,000)                                                                       |
| 2025/26 | SUSTech Outstanding Communist Youth League Cadre                                                                                                         |
| 2026    | Second Prize, CCF Service Computing Innovation Competition / Sangfor CoStrict AI-Native Innovation Competition; GeoAgent (team project)                  |

Leadership & Service

**President, Student Union** | **Chair, Student Development Committee:** lead organizational planning, team coordination, recruitment, student activities, and operating procedures.

**Head, Zhiren College Peer Mentor Team:** lead 15 mentors serving 231 incoming students across 5 classes; organize onboarding, orientation, and ongoing support.

**Collective honors:** SUSTech Outstanding Student Union; First Prize, Zhiren College Outstanding Student Organization (2025–2026).

**Volunteer service:** **87.17 verified hours**, including campus events and accessibility initiatives.

**Other roles:** Turing Class President; 3D Manufacturing Club Vice President; competitive programming, supercomputing, rowing, emcee, and exhibition hall docent teams.